Date	4/28/2026
Group	49
Member	Charan Sriram
Member	Jai Patel
Homework	Project
Due	4/28/2026

Project Description:

This project is a command line interface project called TPC-DS Item Rental Management System that connects to Maria DB with data from the TPC-DS Benchmark Dataset. This project allows staff users to manage item inventory, customer accounts, active rentals, waitlists, and rental history. It also allows users to add items, add customers, rent items, return items, grant rental extensions, search through any table using different types of filters, and manage a waitlist for items that are out of stock. The goal of this project is to make a python file that runs SQL queries in order to interact with the data, using the command line, according to your needs.

What Works:

- Adding new items to the item table
- Adding new customers with address and name information
- Editing existing customer information including name and address
- Renting an item to a customer and putting an automatic 14 day due date
- Returning a rented item, moving it to rental history, and putting the return date to the date the rented item was returned
- Granting a 14 day extension to the due date of an active rental
- Adding a customer to the waitlist when an item is not there
- Updating the waitlist when an item becomes available
- Returning the number of rows in rental with a certain item's id
- Returning a customer's place in line from the waitlist for a certain item
- Returning the number of rows in the waitlist for a certain item.
- Searching and filtering items by ID, name, brand, category, manufacturer, price range, and start year
- Searching and filtering customers by ID, name, email, and address
- Filtering and searching rentals by customer ID, item ID, rental date, and due date
- Searching and filtering rental history by item ID, customer ID, rental date, return date, and due date

- Searching and filtering the waitlist by customer ID, itemID, and place in line
- Saving all changes to the database
- Closing the database connection cleanly

What Does Not Works:

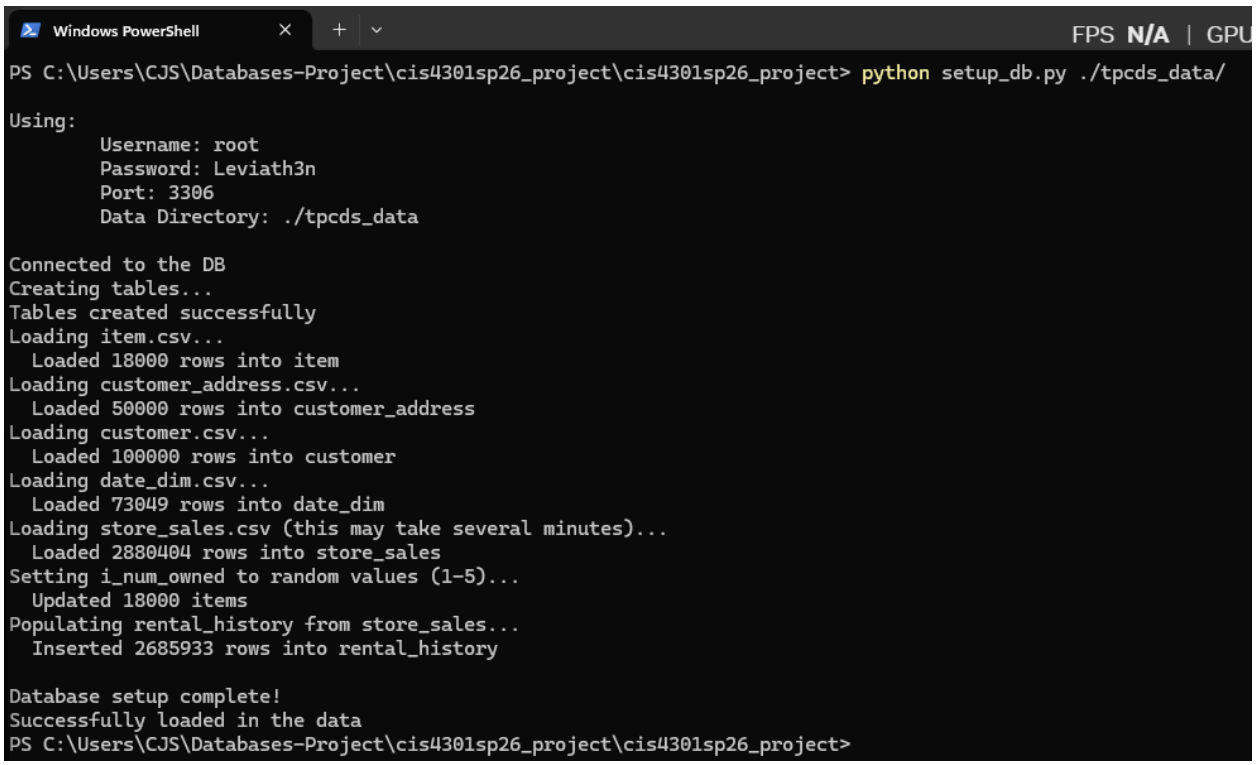
- We aren't aware of anything not working at this time

Known Bugs:

- No known bugs

The screenshots defined in Section 11.2. :

01_setup_db.png:



02_rent_item.png:

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 1

Enter Item ID: SSDEMOITEM787878
Enter Customer ID: SSDEMOCUSTOMER78
Successfully rented item
```

03_return_item.png:

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 2

Enter Item ID: SSDEMOITEM787878
Enter Customer ID: SSDEMOCUSTOMER78
Successfully returned the item
```

04_grant_extension.png:

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 3

Enter Item ID: SSDEMOITEM787878
Enter Customer ID: SSDEMOCUSTOMER78
Successfully granted extension

What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 3

Enter Item ID: SSDEMOITEM787878
Enter Customer ID: SSDEMOCUSTOMER78
The customer already has an extension and may not be granted another one
```

05_search_item.png:

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 4

Which table would you like to search?
1. Item
2. Customer
3. Rental
4. Rental History
5. Waitlist
6. Cancel

Choice: 1

Would you like to use patterns to search String attributes? (Y/N): N
Which attribute would you like to filter?
1. Item ID
2. Product Name
3. Brand
4. Manufacturer
5. Category
6. Min Price
7. Max Price
8. Min Start Year
9. Max Start Year
10. Continue
11. Cancel

Choice: 2

Product Name: SS Demo Product

Current Filters:
--------------------
Product Name: SS Demo Product
--------------------
```

```
Which attribute would you like to filter?
1. Item ID
2. Product Name
3. Brand
4. Manufacturer
5. Category
6. Min Price
7. Max Price
8. Min Start Year
9. Max Start Year
10. Continue
11. Cancel

Choice: 4

Manufacturer: Sriram Inc

Current Filters:
--------------------
Product Name: SS Demo Product
Manufacturer: Sriram Inc
--------------------

Which attribute would you like to filter?
1. Item ID
2. Product Name
3. Brand
4. Manufacturer
5. Category
6. Min Price
7. Max Price
8. Min Start Year
9. Max Start Year
10. Continue
11. Cancel

Choice: 10


Current Filters:
--------------------
Product Name: SS Demo Product
Manufacturer: Sriram Inc
--------------------

--------------------
Item ID: SSDEMOITEM787878
Product Name: SS Demo Product
Brand: Patel Inc
Category: Electronics
Manufacturer: Sriram Inc
Current Price: 5.99
Start Year: 2010
Total number of copies owned: 14
--------------------

Found 1 item.
```

06_search_customer.png:

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 4

Which table would you like to search?
1. Item
2. Customer
3. Rental
4. Rental History
5. Waitlist
6. Cancel

Choice: 2

Would you like to use patterns to search String attributes? (Y/N): Y
Which attribute would you like to filter?
1. Customer ID
2. Name
3. Address
4. Email
5. Continue
6. Cancel

Choice: 1

Customer ID: %78

Current Filters:
--------------------
Customer ID: %78
--------------------
```

```
Which attribute would you like to filter?
1. Customer ID
2. Name
3. Address
4. Email
5. Continue
6. Cancel

Choice: 2

Name: %ames

Current Filters:
--------------------
Customer ID: %78
Name: %ames
--------------------

Which attribute would you like to filter?
1. Customer ID
2. Name
3. Address
4. Email
5. Continue
6. Cancel

Choice: 5


Current Filters:
--------------------
Customer ID: %78
Name: %ames
--------------------


--------------------
Customer ID: SSDEMOCUSTOMER78
Name: LeBron James
Address: 23 Goat Circle, Los Angeles, CA 90015
Email: GloriousKingJames@Lakers.com
--------------------

Found 1 customer.
```

07_search_rental.png:

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 4

Which table would you like to search?
1. Item
2. Customer
3. Rental
4. Rental History
5. Waitlist
6. Cancel

Choice: 3

Which attribute would you like to filter?
1. Item ID
2. Customer ID
3. Min Rental Date
4. Max Rental Date
5. Min Due Date
6. Max Due Date
7. Continue
8. Cancel

Choice: 3

Min Rental Date (YYYY-MM-DD): 2026-04-27

Current Filters:
--------------------
Min Rental Date: 2026-04-27
--------------------
```

```
Which attribute would you like to filter?
1. Item ID
2. Customer ID
3. Min Rental Date
4. Max Rental Date
5. Min Due Date
6. Max Due Date
7. Continue
8. Cancel

Choice: 4

Max Rental Date (YYYY-MM-DD): 2026-04-29

Current Filters:
--------------------
Min Rental Date: 2026-04-27
Max Rental Date: 2026-04-29
--------------------

Which attribute would you like to filter?
1. Item ID
2. Customer ID
3. Min Rental Date
4. Max Rental Date
5. Min Due Date
6. Max Due Date
7. Continue
8. Cancel

Choice: 7


Current Filters:
--------------------
Min Rental Date: 2026-04-27
Max Rental Date: 2026-04-29
--------------------

--------------------
Item ID: SSDEMOITEM787878
Customer ID: SSDEMOCUSTOMER78
Rental Date: 2026-04-28
Due Date: 2026-05-12
--------------------

Found 1 rental.
```

08_search_rental_history.png:

```
PS C:\Users\CJS\Databases-Project\cis4301sp26_project\cis4301sp26_project> python main.py
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 4

Which table would you like to search?
1. Item
2. Customer
3. Rental
4. Rental History
5. Waitlist
6. Cancel

Choice: 4

Which attribute would you like to filter?
1. Item ID
2. Customer ID
3. Min Rental Date
4. Max Rental Date
5. Min Due Date
6. Max Due Date
7. Min Return Date
8. Max Return Date
9. Continue
10. Cancel

Choice: 3

Min Rental Date (YYYY-MM-DD): 2002-05-23

Current Filters:
--------------------
Min Rental Date: 2002-05-23
--------------------
```

```
Max Rental Date (YYYY-MM-DD): 2002-05-23

Current Filters:
--------------------
Min Rental Date: 2002-05-23
Max Rental Date: 2002-05-23
--------------------

Which attribute would you like to filter?
1. Item ID
2. Customer ID
3. Min Rental Date
4. Max Rental Date
5. Min Due Date
6. Max Due Date
7. Min Return Date
8. Max Return Date
9. Continue
10. Cancel

Choice: 9


Current Filters:
--------------------
Min Rental Date: 2002-05-23
Max Rental Date: 2002-05-23
--------------------


--------------------
Item ID: AAAAAAAAAABDAAAA
Customer ID: AAAAAAAADGOBBAAA
Rental Date: 2002-05-23
Due Date: 2002-06-06
Return Date: 2002-06-06
--------------------
--------------------
Item ID: AAAAAAAAAACDAAAA
Customer ID: AAAAAAAAMMGBAAAA
Rental Date: 2002-05-23
Due Date: 2002-06-06
Return Date: 2002-06-06
--------------------
```

09_search_waitlist.png:

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 1

Enter Item ID: SSDEMOWAITLISTIT
Enter Customer ID: SSDEMOWAITLISTCU
This item is not available right now.
Would you like to waitlist the Customer (Y/N): Y
The customer is now 1st in line to rent the item

What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 4

Which table would you like to search?
1. Item
2. Customer
3. Rental
4. Rental History
5. Waitlist
6. Cancel

Choice: 5

Which attribute would you like to filter?
1. Item ID
2. Customer ID
3. Min Place in line
4. Max Place in line
5. Continue
6. Cancel

Choice: 1

Item ID: SSDEMOWAITLISTIT

Current Filters:
--------------------
Item ID: SSDEMOWAITLISTIT
--------------------
```

```
Which attribute would you like to filter?
1. Item ID
2. Customer ID
3. Min Place in line
4. Max Place in line
5. Continue
6. Cancel

Choice: 5


Current Filters:
--------------------
Item ID: SSDEMOWAITLISTIT
--------------------


--------------------
Item ID: SSDEMOWAITLISTIT
Customer ID: SSDEMOWAITLISTCU
Place in line: 1
--------------------

Found 1 waitlisted customer.
```

10_add_item.png:

```
PS C:\Users\CJS\Databases-Project\cis4301sp26_project\cis4301sp26_project> python main.py
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 5

Enter Item ID: SSDEMOITEM787878
Enter Product Name: SS Demo Product
Enter Brand: Patel Inc
Enter Category: Electronics
Enter Manufacturer: Sriram Inc
Enter Current Price: 5.99
Enter Start Year: 2010
Enter the number of copies owned: 14
```

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 5

Enter Item ID: SSDEMOITEM787878
An item with that ID already exists.
```

11_add_customer.png:

```
PS C:\Users\CJS\Databases-Project\cis4301sp26_project\cis4301sp26_project> python main.py
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 6

Enter Customer ID: SSDEMOCUSTOMER78
Enter First Name: LeBron
Enter Last Name: James
Enter Email: GloriousKingJames@Lakers.com
Enter Street Number: 23
Enter Street Name: Goat Circle
Enter City: Los Angeles
Enter State: CA
Enter Zip: 90015
```

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 6

Enter Customer ID: SSDEMOCUSTOMER78
A customer with that ID already exists.
```

12_edit_customer.png:

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 7

Customer's Customer ID: SSDEMOCUSTOMER78

Which attribute would you like to edit?
1. Customer ID
2. Name
3. Address
4. Email
5. Save
6. Cancel

Choice: 1

Customer ID: FOURTIMEMVPLEBRO
Current Attributes:
--------------------
Customer ID: FOURTIMEMVPLEBRO
--------------------

Which attribute would you like to edit?
1. Customer ID
2. Name
3. Address
4. Email
5. Save
6. Cancel

Choice: 2

Name: King James
Current Attributes:
--------------------
Customer ID: FOURTIMEMVPLEBRO
Name: King James
--------------------
```

```
Which attribute would you like to edit?
1. Customer ID
2. Name
3. Address
4. Email
5. Save
6. Cancel

Choice: 5

Current Attributes:
--------------------
Customer ID: FOURTIMEMVPLEBRO
Name: King James
--------------------


What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 4

Which table would you like to search?
1. Item
2. Customer
3. Rental
4. Rental History
5. Waitlist
6. Cancel

Choice: 2

Would you like to use patterns to search String attributes? (Y/N): N
```

```
Which attribute would you like to filter?
1. Customer ID
2. Name
3. Address
4. Email
5. Continue
6. Cancel

Choice: 1

Customer ID: FOURTIMEMVPLEBRO

Current Filters:
--------------------
Customer ID: FOURTIMEMVPLEBRO
--------------------

Which attribute would you like to filter?
1. Customer ID
2. Name
3. Address
4. Email
5. Continue
6. Cancel

Choice: 5


Current Filters:
--------------------
Customer ID: FOURTIMEMVPLEBRO
--------------------


--------------------
Customer ID: FOURTIMEMVPLEBRO
Name: King James
Address: 23 Goat Circle, Los Angeles, CA 90015
Email: GloriousKingJames@Lakers.com
--------------------

Found 1 customer.
```